PROJECT MANAGEMENT FOUNDATIONS

R 250917

Project activities and organization

Project management framework

Review questions

Project activities: management and execution

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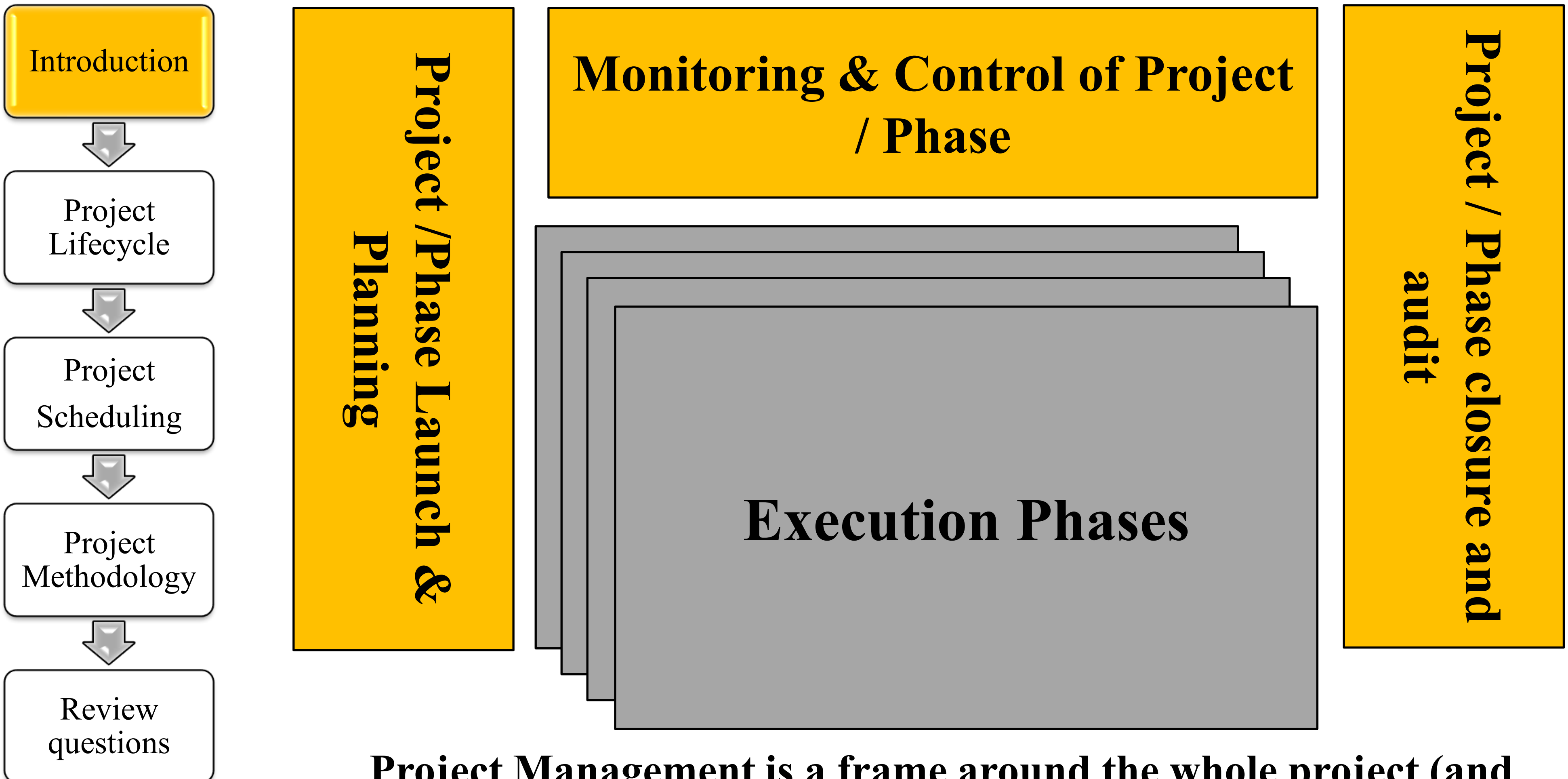
- A project is a non-repeating business process
  - Lasts a pre-defined time period
  - Has a specific outcome e.g., a new system
- **A project orchestrates diverse layers of activity**
  - **Execution: a sequence of phases and/or work packages made of operational activities (e.g., software development)**
  - **Management: a variety of governance activities**

Project activities: management and execution



Project Management is a frame around the whole project (and project phases in larger projects)

Project activities: Management cycle

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Project management is a cyclical process that includes

1. Planning: setting objectives, goals/deadlines and related achievement measures, and approve the plan
2. Monitoring: collecting data on the actual performance and environment changes and notifying alarms as needed
3. Controlling: assessing the progress against plan and setting corrective actions as needed, hence updating the plan

Project activities: Management cycle

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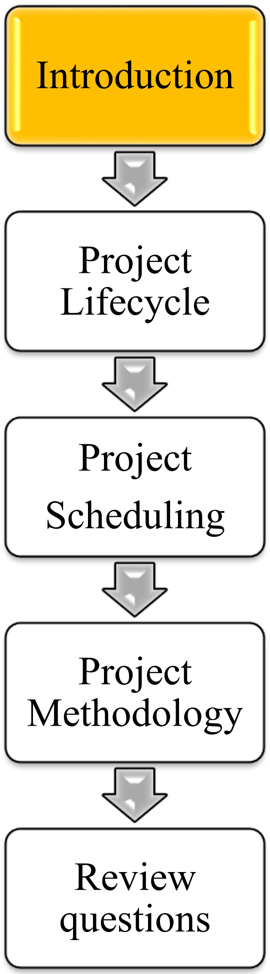
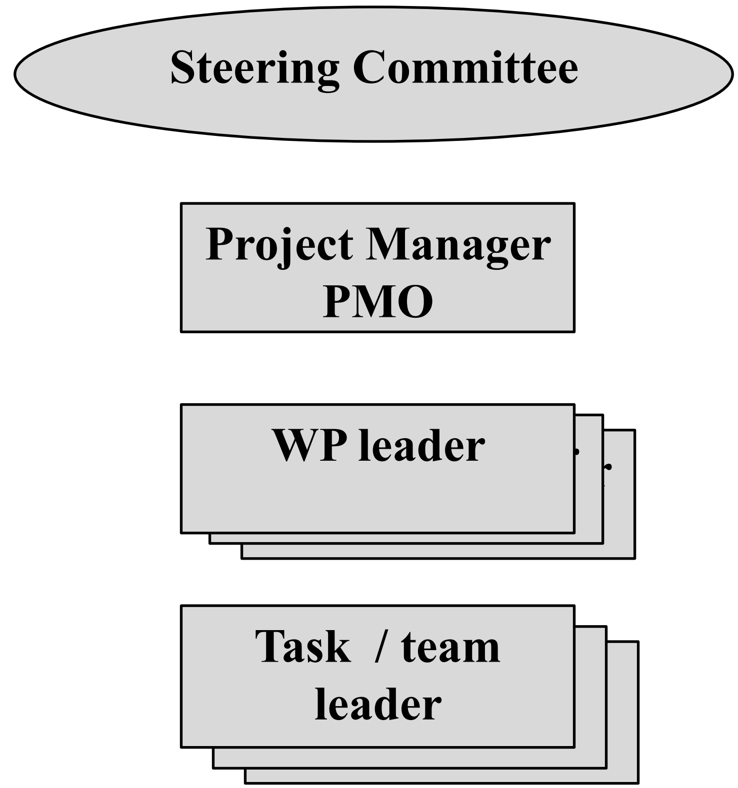
- The frequency and intensity of the management cycle is variable: the more complex the project the tighter the management cycle
- A weekly/monthly cycle is customary, and the frequency of the management cycle typically is:
  - Planning:
    - Whole project: at the project start and at every milestone
    - Project team: monthly /weekly
  - Control:
    - Whole project: monthly
    - Project team: weekly
  - Monitoring: less or more intensive, depending on expected project size, risks and complexity

Project organization



A project typical structure relies on various management roles:

- Project policy /politics: Steering Committee which includes project sponsors and major stakeholders
- Management of the whole project: Project Manager (PM) with the Project Management Office (PMO)
- Management of a project Work Package (WP): WP Leader
- Management of a WP task: Team Leader



Project responsibilities

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Project Management Levels	**Project organization**				
	Steering Committee	**Project Manager**	**Project Management Office**	**Work Package Leader**	**Team Leader**
Planning	D (approves or rejects)	E (overall project)	A (prepares and updates plan)	E (work package)	E (team)
Control	I/D	A (overall project)	E (for the whole project)	E (work package)	E (team)
Monitoring	(I)	I/D	E (reports on progress)	E	E

- **E= Executes**
- **D = Decides**
- **I = Is informed**
- **(I) = Is informed as needed**

- **A= Assists or supports**
- **(A) = Assists as needed / on request**
- **I/D = Is informed and decides if needed**

Project scope and size

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- Digital Projects can address multiple domains
  - Software development / customization / training
  - Hardware and infrastructure design/installation/ acquisition
  - Change in the business and human resources
- The size of Digital Projects may be
  - Micro : up to 15 person months
  - Small : up to 50 person months
  - Large : up to 200 person months
  - Very large : up to 1,000 person months
  - Giant : over 1,000 person months

Programs

- Large projects addressing multiple domains are often called **Programs**
- Programs are often managed at the 2 levels:
  - the overall Program
  - the Projects which address a part of the program
- E.G.: A New Banking Program includes several projects
  1. Overall Program (Program Office)
  2. Marketing and Communication (Marketing)
  3. Branch staff training (Human Resource and Organization Change)
  4. New Customer’s AI Apps (Software)
  5. International Dissemination (Software and Infrastructure)
  6. New Counter System (Software)
  7. New Server Farm (Infrastructure)

Evolution of project methodologies

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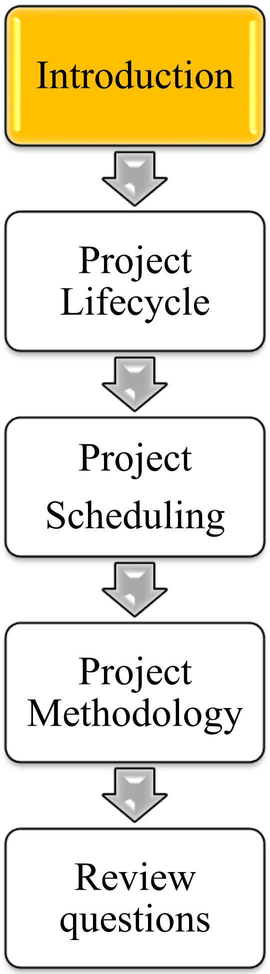
1910: construction management	• Gantt's chart • Work package/work breakdown structure concept
1960-70 (Apollo project): Project Management	• Oriented graph techniques: PERT & CPM • Cleland's text on Project Management
1970-90: Information Systems Project Methodologies	• Analysis /design techniques • Checklists for project activities, documentation, responsibility
1990 - Today: Integrated Project Management	• Project Management Book of Knowledge (PMBOK) • Agile and Scaled Agile paradigms
2020 - Today: AI in the Project	• Generative AI tools for project (AI agents etc.) • AI projects

The Project Management framework



- Project management relies on these models:
  1. Lifecycle: a model of (digital) project phases and approaches to run the lifecycle
  2. Scheduling: a model of project activities and resources
  3. Methodology: a model of the processes for managing the project
- These concepts are illustrated in the following sections

Project Life Cycle

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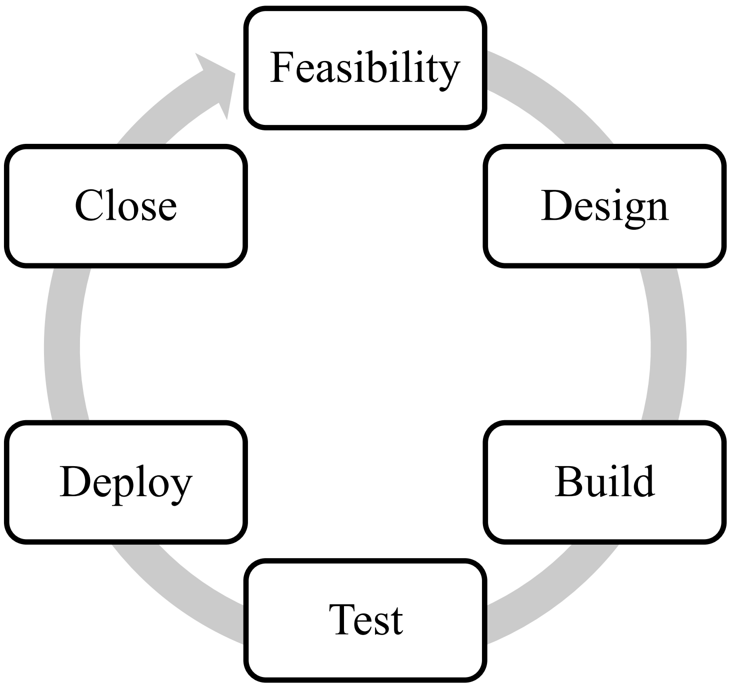

- A project evolves in phases
- The sequence of phases is called life cycle
- A lifecycle can be performed by different approaches, as:
    1. Predictive
    2. Adaptive
    3. Hybrid (a part of the project is adaptive and a part is predictive)

Predictive approaches: Waterfall (Wikipedia's definition)

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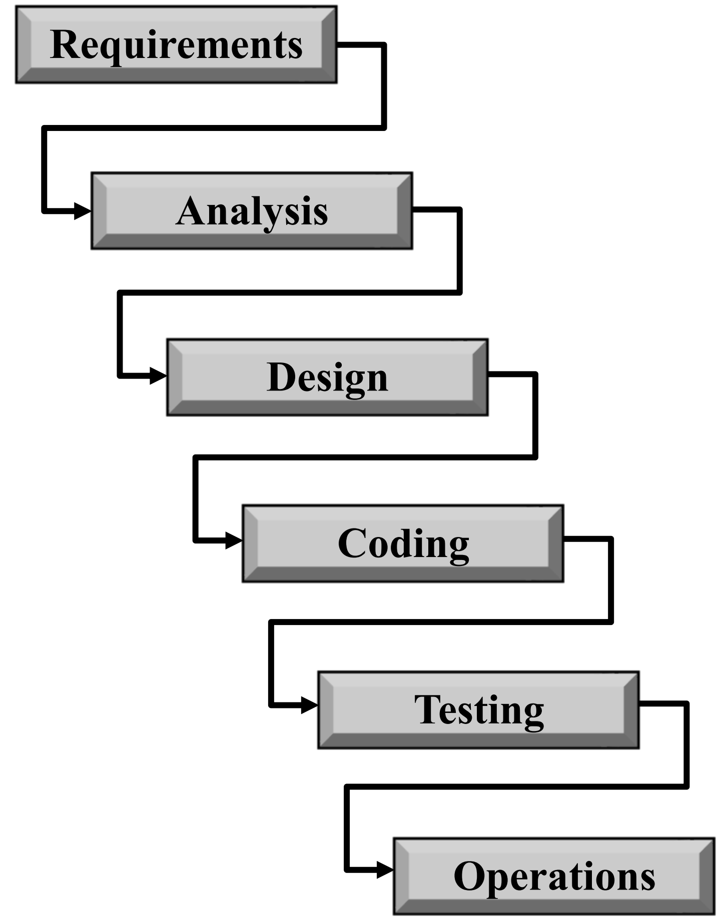
Winston W. Royce (1970) wrote the first waterfall model with standard phases of an IT project:

1. System and software requirements: a product requirements document
2. Analysis: models, schema, and business rules
3. Design: system architecture
4. Coding: development, proving, and integration of software
5. Testing: systematic discovery and debugging of defects
6. Operations: installation, migration, s upport, and maintenance of complete systems



Predictive approaches: Waterfall: comments

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- Waterfall is the most typical predictive approach for IT projects
- In "Waterfall", quality control and spec compliance are a must. Hence:
  - Each project phase should be finished and validated before the subsequent phase can begin
  - No change is allowed during the project
- Waterfall is consistent with major methodologies
  - Capability Maturity Model Integration (CMMI)
  - Project Management Book of Knowledge (PMBOK)

Predictive approaches: Waterfall: comments

- Waterfall fits
  - firm and cogent requirements and complex and interdependent specs, e.g., in telecommunications software
- Waterfall unfits
  - uncertain and loose requirements, where software developers adjust software on user’s feedback
  - application platform customization, since specs and requirements are incorporated in the platform

Adaptive approaches

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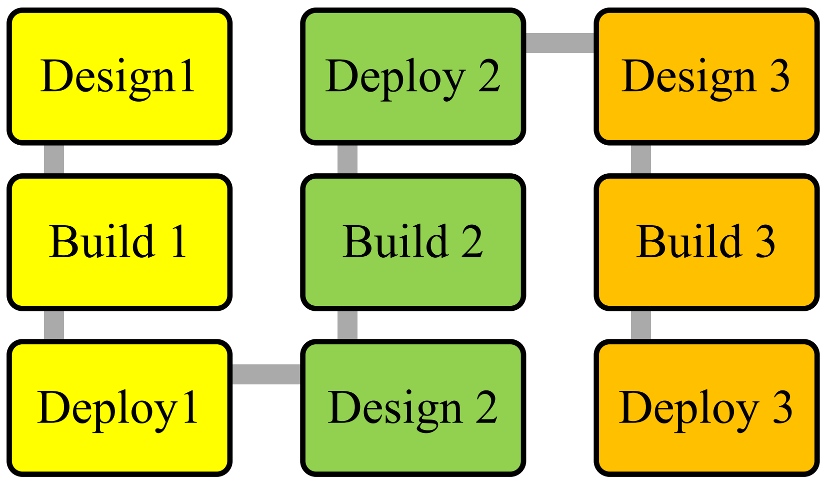
- In adaptive approaches the project is planned and built in multiple stages
- Hence, the project proceeds incrementally
- The Agile approach is the most famous adaptive / incremental approach



Adaptive approaches: Agile

- Agile is a relatively recent concept, that has met a growing success
- Agile targets software development activities where:
  - Requirements concern few functions (even 1)
  - Requirements can be expressed in plain text («user story»)
  - Requirements may be uncertain and an ongoing adjustment to implemented software is convenient
  - An incremental development withe short cycles is welcome
- "Scrum" (2001) is the most popular Agile methodology, conceived for small projects
- In large projects, scaled Agile methodologies have been developed and hybrid approaches are used

Hybrid approaches

- In a hybrid approach, Waterfall and Agile are used concurrently, for instance:
  - Building in a predictive way a product module straight from user requirements to operation
  - Building a second product module incrementally
    - by first building a prototype of the system
    - by a sequence of releases each one realizing a set of features
    - etc.

From project management to systems management

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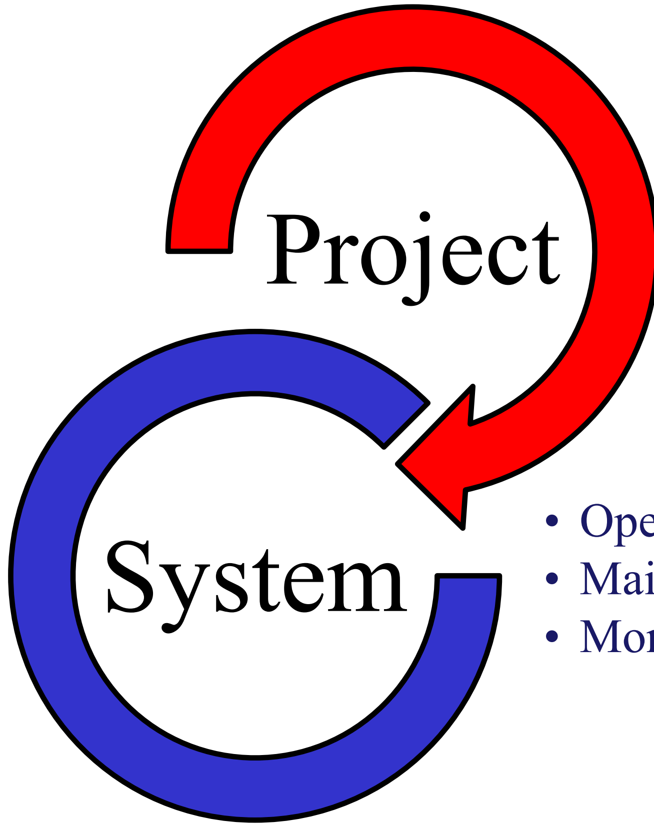

- Feasibility
- Design
- Build
- Test
- Deploy
- Close project

- Operate
- Maintain
- Monitor

Management continue after the project has been closed

From project management to systems management

- Systems continue after the project has been closed
  - They need to be run on premises and/or cloud
  - They should be maintained to:
    - Fit changes in law and/or regulations and/or structure-rules of the enterprise (adaptive maintenance)
    - Fix errors and/or analysis misunderstanding (corrective maintenance)
    - Add new features (improvement)
  - They should be monitored in terms of
    - Service performances (Service Level Management)
    - Satisfaction of users and stakeholders
- Additionally, AI systems should be monitored in terms
    - Quality of results (correctness, bias)
    - Ethic compliance
- In a comprehensive view, project management is integrated by systems management

Project Scheduling

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Scheduling techniques support the project manager in setting and controlling the calendar of project activities.

Scheduling address:

- Activity deadlines
- Activity dependencies
- Resource constrains

Scheduling techniques include:

- Graph techniques: Gantt, PERT, CPM
- Work sequencing methods: Kanban

Project Scheduling: summary

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	Duration	Progress	Dependency	Resources	Ace
GANTT	**Y**	**Y**	**N**	**N**	Simplicity and flexibility Multilevel Planning & Controlling
PERT	**Y**	**Y**	**Y**	**Y**	Multilevel Planning & Controlling Dependencies
CPM	**Y**	**(Y)**	**Y**	**N**	Dependencies Critical Path
KANBAN	***(Y)***	**Y**	***N***	***N***	*Visual communication*

Gantt Chart

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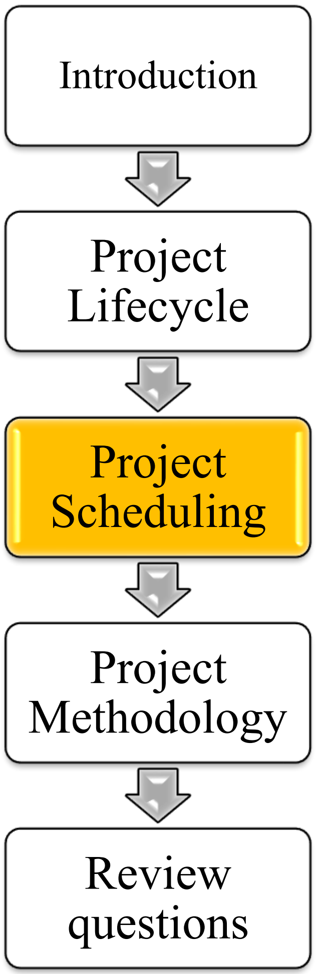
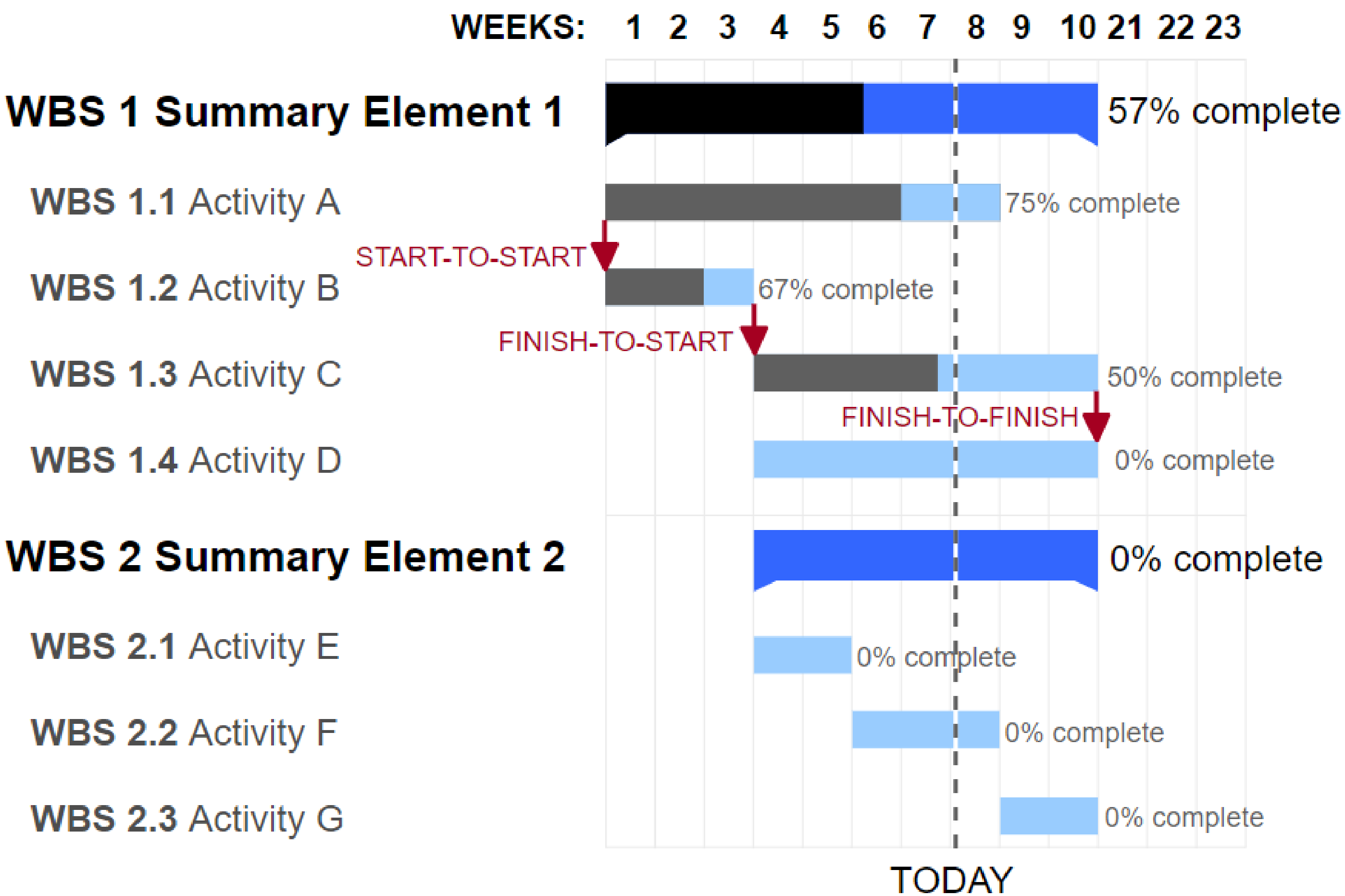
- Devised by Henry Gantt in the 1910s
- Shows start and finish dates of project activities
- Activities show the structure of the project work, and therefore are called Work Breakdown Structure (WBS)
- WBS may be decomposed at multiple levels; hence you can have
  - First level WBS, aka Summary Element
  - Second level WBS, sometimes called task or activity
  - …
  - N-level WBS
- In its simplest form, Gantt chart is a timeline of project
- The Gantt Chart can show also the progress of activities (filled lines)

Gantt Chart: example





Gantt Chart

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- The chart can be augmented by dependencies among WBS
  - Finish to Finish (activities should finish at the same time)
  - Finish to Start (an activity should start only after another activity is finished)
  - Start to Start (activities should start at the same time)
- WBSs can be structured by various criteria: Product, Activity, Geography, Organization, or a mix of these criteria.

Program Evaluation and Review Technique - PERT Part1

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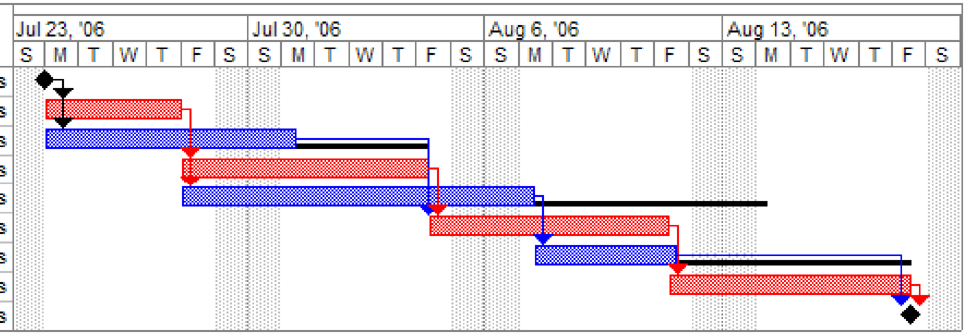
Credit: Wiki

ID	Task Name	Predecessors	Duration
1	Start		0 days
2	a	1	4 days
3	b	1	5.33 days
4	c	2	5.17 days
5	d	2	6.33 days
6	e	3,4	5.17 days
7	f	5	4.5 days
8	g	6	5.17 days
9	Finish	7,8	0 days



- A development of Gantt model
- A statistical tool to model the tasks of a project.
- Developed for the U.S. Navy Special Projects Office in 1957 (U.S. Navy's Polaris nuclear submarine).
- Applications all over industry.
- Commonly used with the Critical Path Method (CPM).

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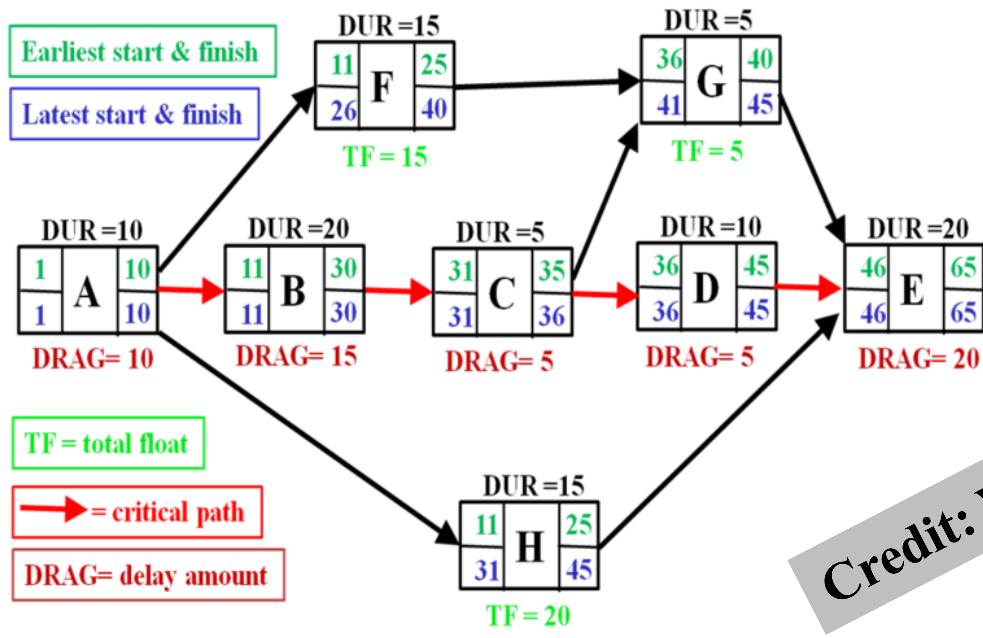
Critical Path Method - CPM

A development of Gantt based on graph theory

Developed in the late 1950s by Morgan R. Walker of DuPont and James E. Kelley Jr. of Remington Rand



Credit: Wiki

TF = total float
Float or slack is the amount of time that a task in a project network can be delayed without causing a delay to subsequent tasks ("free float") or project completion date ("total float").

Critical path drag *is the amount of time that an activity (or constraint on the critical path) is adding to the project duration. Alternatively, it is the maximum time one can shorten the activity before it is no longer on the critical path or before its duration becomes zero.*

Basic Input

- List of project activities
- Duration of each activity
- Dependencies between the activities
- Logical end points (milestones or deliverable items)

Project scheduling technology: Microsoft Project

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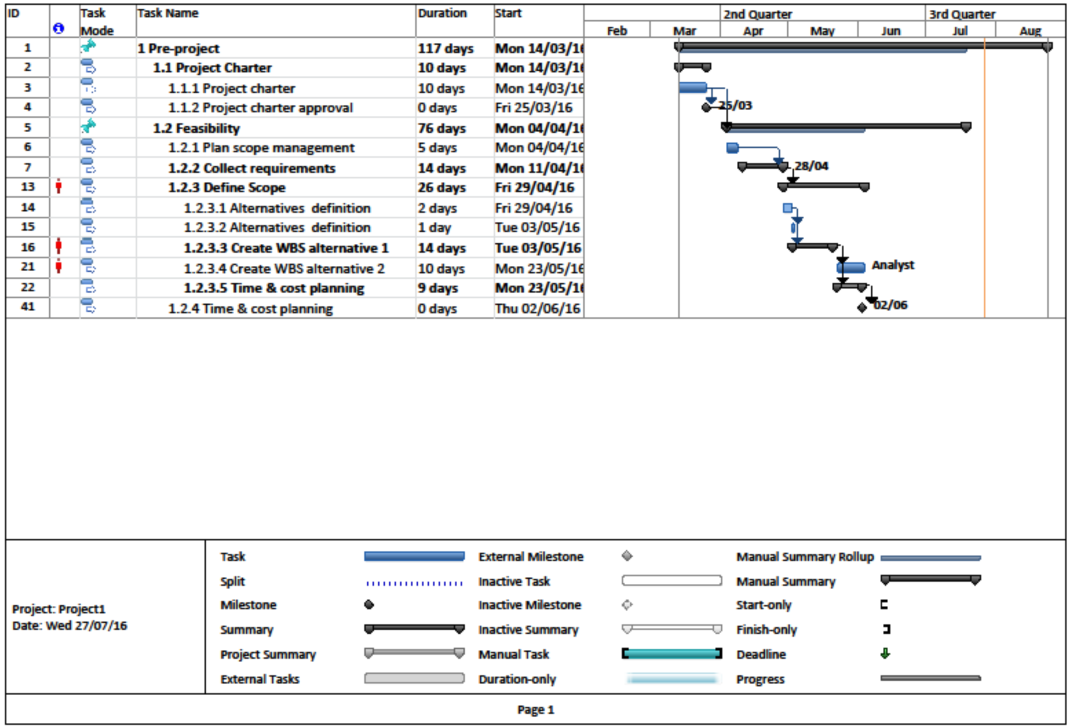

- A project management software by Microsoft
- Assists a project manager in developing a plan, assigning resources to tasks, tracking progress, managing the budget, and analyzing workloads.
- Dominant PC-based project management software.

Project scheduling : KANBAN (credit: wiki)

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- In Kanban work items are visualized to give participants a view of progress and process, from start to finish, usually by a kanban board.
- Work items are visualized to give participants a view of progress and process, from start to finish.
- Dependencies among activities are not modeled

Layers of projects

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- From a modeling perspective a project can be seen as a layered process
- A layer is a domain of knowledge and management which is largely self-contained
- Project layers include:
  1. Overall project structure and project knowledge areas
  2. Project frameworks
  3. Software development processes
  4. Software Engineering Techniques and Tools

Layers of projects

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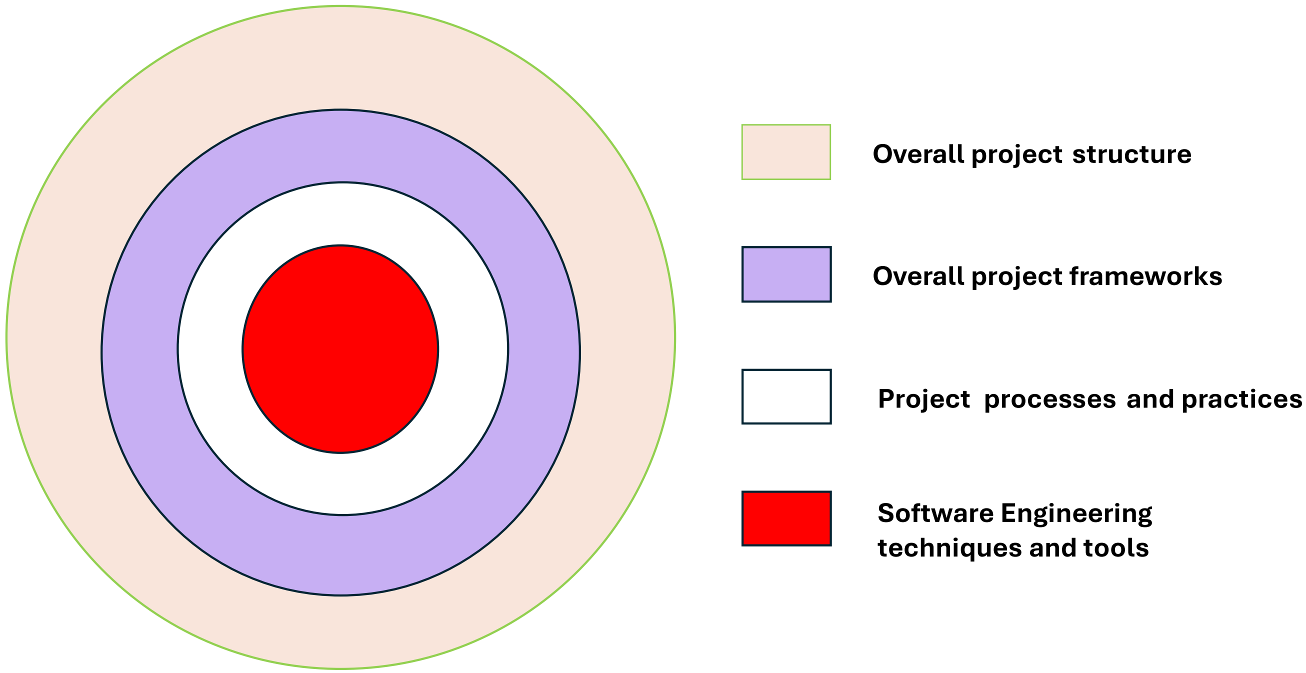

- Each layer can be implemented in an Agile or a Predictive paradigm
- A project can combine Agile and Predictive paradigms at different layers, thus resulting in a hybrid methodology

Layers of projects

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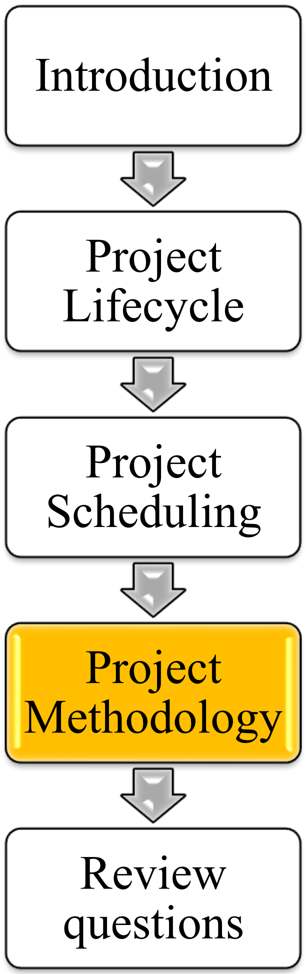
PROJECT LAYER	AGILE PROJECT ANAGEMENT	PREDICTIVE PROJECT MANAGEMENT
Overall project structure and project knowledge areas	Scaled Agile Frameworks (e.g., SAFe, LeSS, Disciplined Agile)	Project Management Frameworks (e.g., PMBOK, PRINCE2)
Project frameworks	Agile Frameworks (Scrum, Kanban, XP, etc.)	Phased project frameworks (Waterfall, Linear, Iterative, etc.)
Software development processes	Agile Processes and Practices (e.g., Sprint, Backlog, Iteration)	Software Development Processes and Practices
Software Engineering Techniques and Tools	Agile Techniques and Tools (e.g., User Story, MVP, Pair Programming, TDD, Continuous Integration)	Abstract Modeling Languages (e.g., BPMN, UML), Coding, Testing, Maintenance tools

Project Methodology



- Project methodologies assist project managers in:
  - Identifying the areas of project, as scheduling, cost, quality, communications, etc.
  - Managing such areas: how to plan, control, monitor each area
  - Choosing techniques and tools to be used
  - Defining project artifacts
  - Scheduling
  - Selecting the appropriate project life cycle models

Project methodology: comments

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- A large project cannot survive without enforcing a project methodology
- In larger projects, people do not even know each other and may work in different countries
- The wider the project scope and the larger project size the more critical project management is
- According to a survey on 477 projects, Hybrid and Agile project management significantly increases success for stakeholders (Gemino, 2021).

Project Methodology: Architecture

- Generally, project methodologies are structured by knowledge area (aka activity domain) e.g.
  - Cost management
  - Risk management
  - Stakeholder management
  - Progress measurement
  - Etc.

Project Methodology: Contents

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Methodologies typically describe, for each project area or domain, :

- Activity: What shall be made
  - A project contains hundreds of activities of different kind
  - Methodology gives a checklist of standard activities and the criteria to customize them to each project
- Actor: Who is involved in an activity
  - E.g. Team leader and/or project manager
- Technique: How an activity is to be executed
  - E.g.: Project planning activity by PERT
- Tool: implements or technology to be used for a technique
  - E.g.: MS Project for PERT
- Deliverable: Outcome of an activity
  - It can be a product (e.g.: software) or a report or artifact
  - Methodology describes the standard template of the deliverable

Project Methodology: Contents

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- Methodologies differ in the number of areas, in their respective focus and detail
- Famous project methodologies are
  - PMBOK (Project Management Book of Knowledge): a methodology guide issued the Project Management Institute (PMI)
  - PRINCE 2 (PRojects IN Controlled Environments) is a structured project management method and certification program, developed as a UK government standard for information systems projects.

Project Methodology: PMBOK

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- PMBOK was conceived as a general methodology to support the project manager in the project "knowledge areas" (= management domains) such as Scoping, Quality, etc.
- In its original versions, PMBOK was
  - a guide for any kind of project (digital, construction, research ..)
  - neutral on project life cycle and approaches
- PMBOK has been continuously updated and now is edition 7

Project Methodologies Practices

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Practices in PRINCE 2

1. Business Case
2. Organizing
3. Quality
4. Plans
5. Risk
6. Issues
7. Progress

Knowledge Areas in PMBOK 6

1. Scope
2. Schedule
3. Cost
4. Quality
5. HR/ resources
6. Communications
7. Risk
8. Procurement
9. Stakeholder
10. Integration

Performance Domains in PMBOK 7

1. Team
2. Development approach and life cycle
3. Planning
4. Project work
5. Delivery
6. Measurement
7. Uncertainty

Research Project Methodologies

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- Research projects develop base knowledge or innovative product/services, e.g.:
  - Wearable devices for disabled people
  - Smart tiles
  - Smart city systems
- Given their innovation purpose, such projects
  - Have a long duration, 3-5 years
  - Are funded by governments
  - Involve a wide set of research bodies and companies
  - Have their own project management paradigm
- We here summarize the H2020 approach

Research Project Methodologies: H2020 proposal documentation

- H2020
  - Overall charter
    - General information
    - Participants
    - Budget
    - Ethics
  - Call-specific questions
    - Excellence (= Achievements of the project)
    - Impact (social, business)
    - Implementation (=structure of the project)
      - PERT & GANTT
      - Description of Work Packages
      - Description of Project Actors

- The H2020 framework is a template to describe a <u>research project</u> The project is based on Work Packages, i.e. assignments aimed to deliver a project outcome

Review questions

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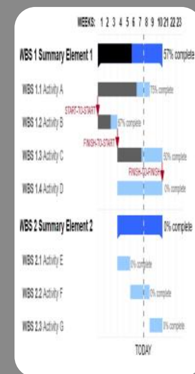
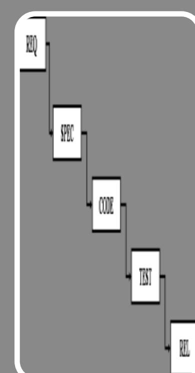
- Review questions address the lecture key points
- The subsequent questions refer to the key concepts of the Introduction



Project scheduling

- GANTT chart
- PERT/CPM



IT Project lifecycle

- Waterfall
- Fit-Gap
- Agile/SCRUM

Review questions: Project Scheduling Milano case study

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- Milan is considered a leading alpha global city, with strengths in the field of the art, commerce, design, education, entertainment, fashion, finance, healthcare, media, services, research and tourism.
- It has the third-largest economy among European cities after Paris and London, but the fastest in growth among the three, and is the wealthiest among European non-capital cities.

Review questions: Project Scheduling
Milano case study

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- The city of Milano plans a new metro line.
- A metro line in Milano
  - Is typically 30 kilometers
  - Is served by around 30 stations, with about 5 stations shared with other metro lines

Review questions: Project Scheduling Milano case study

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WBS ELEMENTS

1. General design
2. Construction work in various locations
3. Procurement and test of metro rail and cars
4. Metro management systems
5. General test

WBS DURATION

- The project lasts 5 years
- Test WBS lasts 1 year, and it is to be the last project phase
- General design WBS lasts1 year

WBS DEPENDENCIES

- Finish to start:
  - Test WBS can start when all other WBSs have been finished
  - General Design WBS should be finished before other project WBSs can start
- General Design and Test can overlap

Review questions: Project Scheduling
Milano case study

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WBS Summary element	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
1. General design					
2. Construction work (various locations)					
3. Procurement and test of metro rail and cars					
4. Metro management systems					
5. General test					

- The diagram shows the initial plan, called baseline
- The diagram is still aggregate: second level WBSs are required to understand how WBS 2, 3, 4 overlap

Review questions: Project Scheduling
Milano case study

WBS Summary element	Year 1	Year 2	Year 3	Year 4	Year 5
1. General design					
2. Construction work (various locations)					
3. Procurement and test of metro rail and cars					
4. Metro management systems					
5. General test					

- The baseline plan decomposes summary WBSs on 3 blocks according to an activity-based decomposition
  - Block 1: design (= requirements and specifications)
  - Block 2: system building to which the main products belong, namely Construction, Rails and Cars, Metro systems
  - Block 3: usage test and roll-out

Review questions: **IT** Project Lifecycle
Prada exercise (imaginative)

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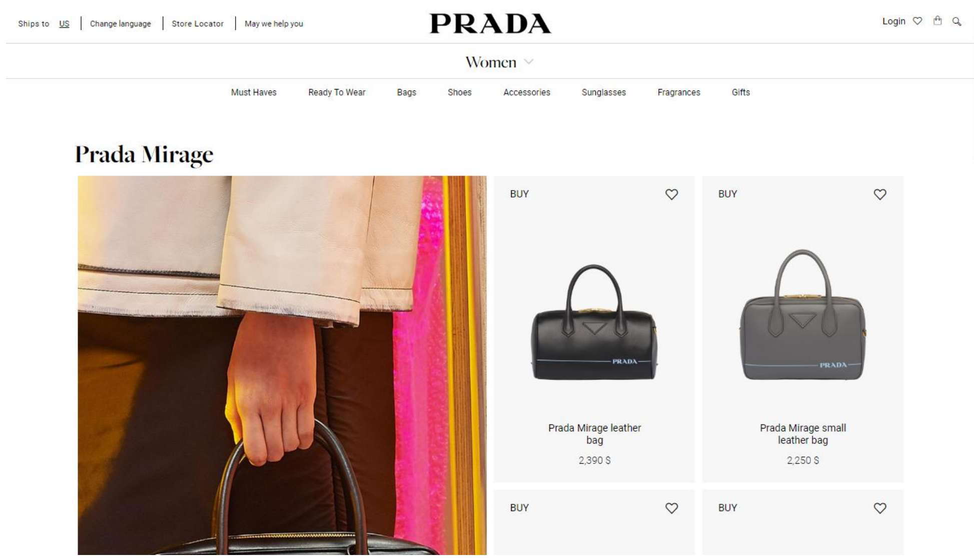

- Prada (4 billion USD income), the famous Italian fashion company, plans to **add** a new mobile interface to its app by which consumers surf Prada's items and shops
- The project is not very large and implies a continuous interaction with users to adjust requirements
- Explain which lifecycle/s can fit that project

Review questions: **IT** Project Lifecycle Prada exercise (imaginative)

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- In Prada project:
  - Requirements concern few functions and a relatively limited budget
  - Requirements are uncertain and an ongoing adjustment to implemented software is convenient
  - A fast development track is possible with small and expert teams
- Hence Agile/ Scrum fits the development of the software product
- Local version (Chinese, Japanese, English, etc.) may be managed by a product backlog

Review questions: <u>IT</u> Project Lifecycle
Ferrero exercise (imaginative)

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Project Methodology

Review questions

- Ferrero an Italian food corporation (10 Billion USD) operates in 55 countries and runs 23 plants
- The ERP project
  - shall support operations across the world
  - shall customize and integrate a commercial application platform.
- Explain which lifecycle/s can fit that project

Review questions: **IT** Project Lifecycle
Ferrero exercise (imaginative)

Introduction

Project Lifecycle

Project Scheduling

Project Methodology

Review questions

- Ferrero project is based on the **customization** of a commercial application platform for ERP
- Hence, a fit-gap analysis /spiral approach fits the software project
- On the other side, the Ferrero project requires a structured and predictive project management because of:
  - Large size of the project – a multimillionaire budget
  - Multiple releases with multiple release teams (e.g. Italy, Germany, China, USA,..)
  - Multiple versions, since a Chinese customization may differ from the American one, due to the different account rules and regulations.